

William Valentine

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EDUCATION

Rose-Hulman Institute of Technology

Terre Haute, IN

Bachelor of Science in Computer Science; Minors in Cognitive Science, Mathematics, Imaging

May 2027

- **Relevant Coursework:** Human-Robot Interaction, Cognitive Psychology, Philosophy of Mind, Computer Vision, Deep Learning, Machine Learning, Linear Algebra, Artificial Intelligence, Deep Reinforcement Learning, Probability with Applications in Statistics

RESEARCH EXPERIENCE

Robotics Institute Summer Scholar (RISS)

Summer 2026 – Present

Carnegie Mellon University

Pittsburgh, PA

- Worked alongside Dr. Aaron Steinfeld to explore social navigation and transportation problems for blind people involving both delivery robots and public buses.
- Work supported by the Safety21 program

Research Intern

Summer 2025 – Present

Indiana University

Bloomington, IN

- Worked alongside Dr. David Crandall and Dr. Selma Šabanović to create novel social robots, including QTrobot and Baxter, for interacting with older adults
- Analyzed large-scale user studies with cross-cultural groups of older adults

REU Intern

Summer 2024 – Winter 2024

University of Nevada, Reno

Reno, NV

- Worked alongside Dr. David Feil-Seifer and Dr. Emily Hand to create one of the first systems for detecting human comfort and discomfort
- Used deep-learning powered pose estimation, valence and arousal, audio transcription, and textual sentiment analysis to model human behavior

Undergraduate Researcher

Fall 2023 – Present

Rose-Hulman Institute of Technology

Terre Haute, IN

- Worked alongside Dr. Michael Wollowski to explore how real-time video-based behavior analysis could be used to create robots with adaptive behavior
- Worked alongside Dr. Michael Wollowski to introduce a system and benchmark for resolving confusing parts of commands given to robots using images
- Worked alongside Dr. Lixing Song to address alignment issues caused by learning-based point cloud alignment methods for self-driving vehicles
- Successfully created an algorithm for aligning point clouds in complex, uncertain environments

VORTEX Member

Summer 2025

Vanderbilt University

Nashville, TN

- Invited to join the first cohort of a summer program focused on research ethics and research writing

EXPERIENCE

Grader and TA

Spring 2024 – Present

CSSE Department, Rose-Hulman Institute of Technology

Terre Haute, IN

- Graded for Machine Learning, Deep Learning, Intro to Software Development, Web Development, Programming Language Concepts, and Data Structures and Algorithms
- Served as Head TA for Programming Language Concepts in Winter 2024 and Winter 2025
- Created and designed an automatic grading system utilizing Python

PUBLICATIONS

Valentine, W., Šabanović, S., Crandall, D., & Khoo, W. (2026). BehaviorKit: A Multi-modal Real-time Behavior Analysis Library for Robots. *HRI 2026 Short Paper*. Acceptance Rate: 38%.

Valentine, W., Wollowski, M. (2025). Assessment of the capabilities of multimodal large language models in locating and resolving ambiguities during human-robot teaming. *AHFE 2025*.

Valentine, W., Webb, M., Collum, C., Feil-Seifer, D., & Hand, E. (2024). HCC: An explainable framework for classifying discomfort from video. *ISVC 2024*. Oral: 45.1%.

Valentine, W., Šabanović, S., Crandall, D., & Khoo, W. (2026). From Long-Term HRI to Dementia Care: Scoping Personalization Approaches. *HRI 2026 Workshop: LEAP-HRI*.

Song, L., **Valentine, W.**, Yang, Q., Wang, H., Fang, H., & Liu, Y. (2024). BB-Align: A lightweight pose recovery framework for vehicle-to-vehicle cooperative perception. *ICDCS 2024*. Acceptance Rate: 21.9%.

Connor, B., **Valentine, W.**, Hsu, L., Kamino, W., Šabanović, S., Crandall, D., & Khoo, W. (2026). Ghost in the Machine: Exploring Designed Fear in HRI. *HRI 2026 LBR*.

Khoo, W., Hsu, L., **Valentine, W.**, Connor, B., Nil, T., Khoo, A., Kamino, W., Šabanović, S., Crandall, D. (2026). Meet IRIS: An Interactive Robotic Experience for Exploring Life Purpose. *HRI 2026 Demo*.

PUBLICATIONS UNDER REVIEW OR REVISION

Valentine, W., Kamino, W., Hsu, L., Šabanović, S., Crandall, D., & Khoo, W. (2026). Novel approach to in-the-wild audio-visual datasets for emotion recognition. *Computer Vision Conference*.

Valentine, W., Wang, Y., Kamino, W., Sato, H., Swaminathan, M., Noguchi, A., Ramirez, F., Hirata, Y., Oh, J., Nagata, S., Tsui, K., Hsu, L., Šabanović, S., Crandall, D., & Khoo, W. (2026). Large-scale cross-cultural user study of social robots interacting with older adults. *Robotics Journal*.

HONORS

Rose-Hulman Institute of Technology <i>Rose Research Fellow and Mentor; Nominated for CSSE TA of the Year</i> - CSSE Student Advisory Board Member (2025-2027) - Academic Computing Implementation Committee Student Representative	Terre Haute, IN
Goldwater Scholar	2026 – 2027
Special Recognition for Outstanding Reviews <i>HRI 2026</i>	2026

SERVICE

Reviewer HRI 2026, CHI 2026, THRI, AI Magazine, AIES

ICDCS Student Volunteer

Associate Managing Editor, CMU RISS Working Papers Journal 2026

GRANTS

National Science Foundation <i>Conference Travel Award, \$500</i>	Alexandria, VA <i>Summer 2024</i>
Rose-Hulman Institute of Technology <i>Rose Research Fellowship and Departmental Grants</i> \$500; \$1000; \$2000; \$1000	Terre Haute, IN <i>Spring 2024, Fall 2024, Spring 2025</i>

SKILLS

Programming Languages: JavaScript, Python, Java, C, Assembly, Scheme

Systems: Windows, Macintosh, Linux